Biohybrid Nanostructures - Wdh 1

Natural Biohybrid Materials

Motivation: Degradable plastics, Optoelectronics, Energy Storage, Medical Engineering

- Biomolecules (Proteins, DNA):
  (C, H, N, O, S)
  - → defined 3D structure
  - → atomically precise
  - → soft (elastic / stretchable)
  - → consist of readily available elements
  - → can be repaired / regenerated
- Inorganic Component:
  - → eg: calcium phosphate, calcium carbonate
  - → mineralized within the organic matrix
  - → hard (stiff, rigid)

⇒ Mostly structural materials, mechanical properties are important

Artificial Biohybrid Materials

"Biohybrid materials consist of bio components, such as biomolecules, cells, and tissues, and other components such as synthetic polymers, ceramics, metals, and metal oxides."

- Biomolecules (Proteins, DNA):
  - → defined 3D Structure, Atomically precise → easy to produce → limited functionality
  - → easy to functionalize / redesign → biocompatible & degradable → limited stability
- Inorganic Component:
  - → eg: nanoparticles, platelets, organometallic complexes
  - → properties eg: optical, electrical, magnetic, catalytic
  - → high thermal and chemical stability
  - → Nanoparticle are not atomically precise → synthesis is hard
  - → Assembly / Structuring can be difficult

⇒ Combine soft and hard component to yield materials with new properties

Biological Macromolecules – Wdh 1

Four Types of Biomacromolecules

	Monomer	Polymer	Cellular Structure
Carbohydrate	Monosaccharide	Starch	Granules in Chloroplast
Nucleic Acid	Nucleotide	Double Helix (DNA)	Chromosome
Protein	Amino Acid	Polypeptide	Intermediate Filaments
Lipid	Fatty Acid	Triglyceride	Adipose Cells

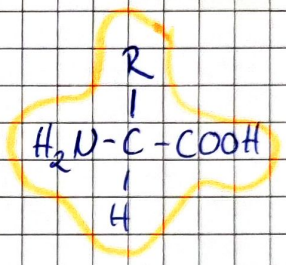
Proteins

Primary Structure > Secondary Structure > Tertiary Structure > Quaternary Str

Primary Structure & Peptide Bonds

Primary Structure: Amino Acid Sequence

Building Blocks: Amino Acids (α-amino acids)



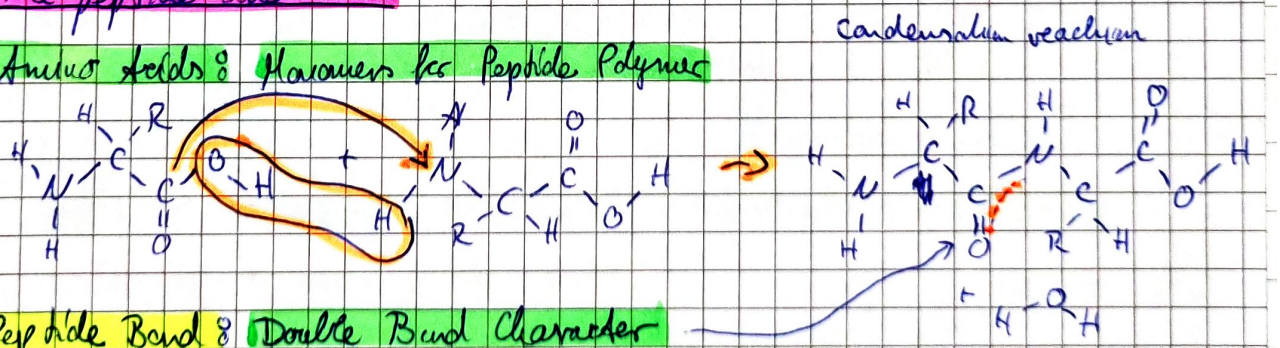
Chirality: α-AA's are found in Nature

21 proteinogenic amino acids:

for exam: learn: Positive, Negative, etc. Lec 01. Slide 16

The peptide bond

Amino Acids: Monomers for Peptide Polymer



Peptide Bond: Double Bond Character

Polar Hydrogen – Nitrogen bonds suitable as hydrogen bond donors

Secondary Structure of Proteins

alpha helix: NH forms H-Bond with C=O of AA (i+4->i) residues apart

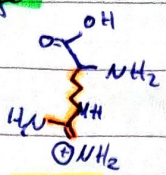
beta sheet: NH forms H-Bond with C=O of AA from opposite strand

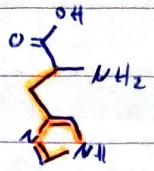
Biologische Nanostrukturen - Wdh 1

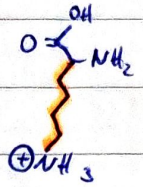
21 proteinogene AA

With Electrical Charge:

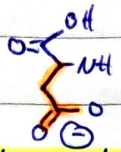
Positive (+):

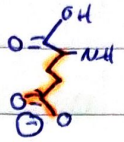

Arginine


Histidine

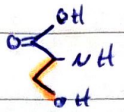

Lysine

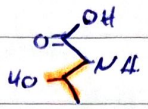
Negative (-):

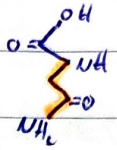

Aspartic Acid

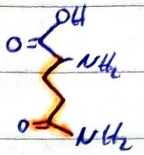

Glutamic Acid

Polar & Uncharged:

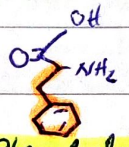

Serine

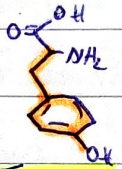

Threonine

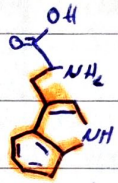

Asparagine


Glutamine

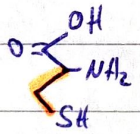
Hydrophobic: Alanine, Valine, Isoleucine, leucine, Methionine

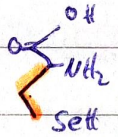

Phenylalanine

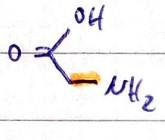

Tyrosine

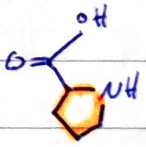

Tryptophan

Special:


Cysteine


Selenocysteine


Glycine


Proline

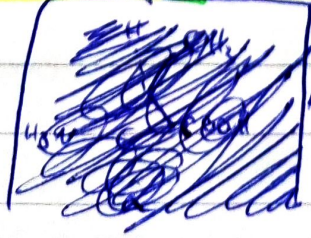
L-peptides in nature!

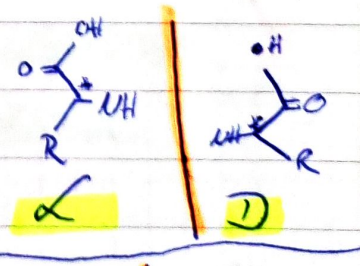
Why are D-peptides interesting for pharmaceutical applications?

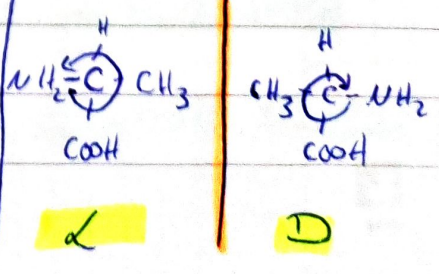
→ Stability & Resistance to Degradation
- L→ Protease Resistance
- L→ more resistant to Hydrolysis

→ Reduced Immunogenicity, since not in nature

→ Enhanced Bioavailability & Extended Activity



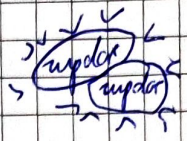




Biophysical Chemistry - Week 1

Hydrophobic effect drives protein folding

- Protein Chain in Water:
  - → unfolded: hydrophobic residues are exposed ☹
  - → folded: hydrophobic residues ~~are~~ form core ☺
- Hydrophobic Molecules:
  - → surrounded by ordered water molecules
  - → aggregation liberates these water molecules → entropic gain



- Protein Folding Contributions:
  - enthalpic: hydrogen bonds, Van der Waals interactions, Electrostatic interactions
  - entropic: hydrophobic effect

unfolded ⇌ folded, ΔG

$\Delta G = \Delta H - T\Delta S < 0$ → fold more than unfold

- Temperature Denaturation with Heat:
  - → decreases enthalpic term, unfolding is energetically favored

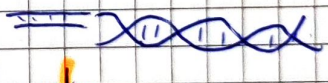
Recombinant Protein Production and Purification

for basic research: eg: enzymes for molecular biology

for industry: eg: vaccines, antibodies

Biosynthetic Pathway for Protein Synthesis

DNA → Protein (A, C, G, T)



- Transcription (DNA → RNA):
  - → piece of DNA is copied to RNA with RNA polymerase
  - → messenger RNA (mRNA) codes for protein

RNA (A, C, G, U)

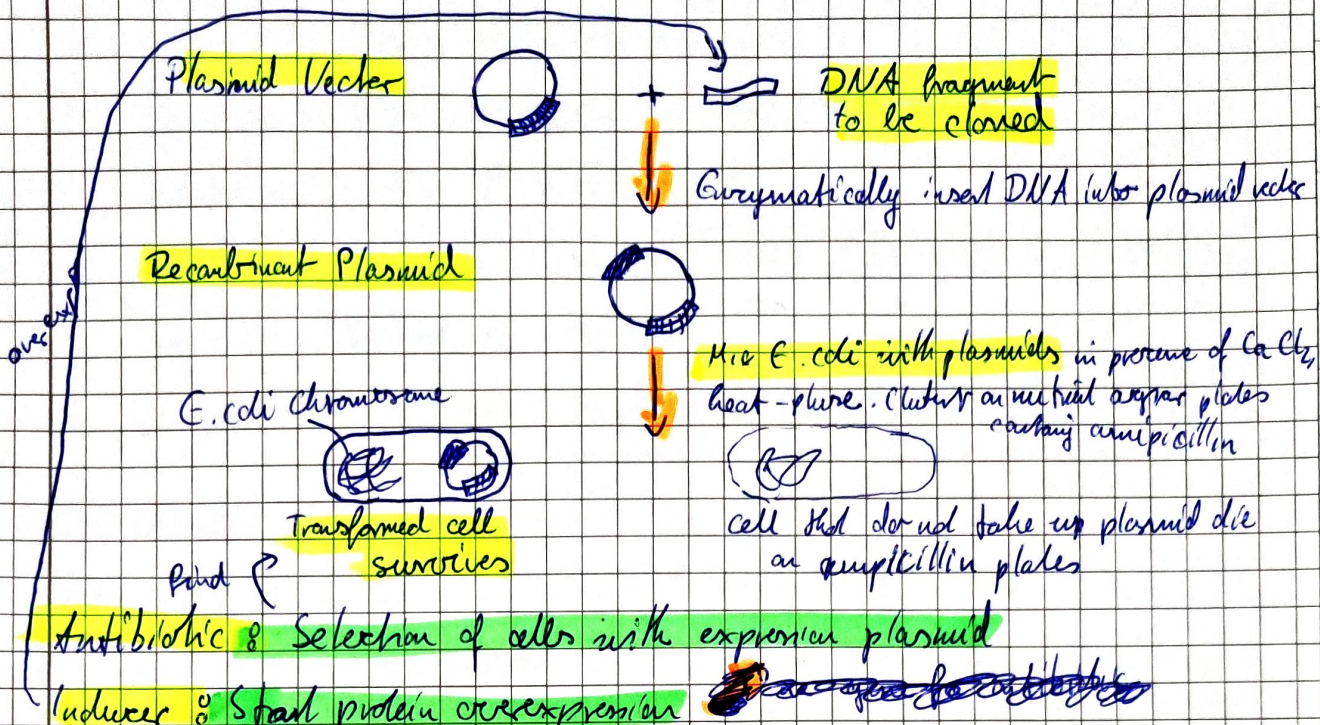
- Translation (RNA → Protein):
  - → mRNA binds to ribosome and is translated to protein via tRNA
  - → tRNA carries one amino acid and has three bases (codons) for recognition

Why is the genetic code degenerate (= several base triplets for the same AA)?

- → Error Minimization: Redundancy provides Buffer, Transcription / Translation Errors,
- → Genetic Robustness: Robust to mutations

Biotyfied Understudies - Wdk 1

Protein Production in E. coli



Advantages:

(+) cheap, low complexity of growth medium

(+) easy introduction of gene of interest, introduction of desired mutations

(+) fast growth of cells

(+) high overexpression of proteins, high yields

Disadvantages:

(-) no eukaryotic folding machinery: eukaryotic proteins can be misfolded

(-) no eukaryotic post-translational modifications: -"- might not be functional

(-) for eukaryotic proteins: yeast cells, insect cells, but more complex & expensive

Richford Neurosciences - Wk 1

Workflow for Recombinant Protein Production

1) Cloning: Gene of interest
2) Expression: Protein gene production started with inducer
3) Harvesting: Cells with protein inside
4) Cell lysis: Cells are disrupted to release protein of interest into solution
5) Purification: Several steps required until protein has desired purity
6) Quality Control: Verification of purity and identity of protein of interest

Cell lysis

Protein produced intracellularly, needs to be released from cells

Methods: Freeze-Thaw Cycles, Osmotic Shock, Physical lysis eg. French press, Microfluidizer, Sonication

Post-treatment with enzymes (DNAse / RNase) usually required

Protein Purification with Chromatography

(blue: positive)
(red: negative)

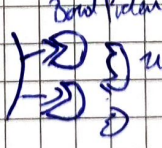
Proteins can be separated according to their properties:

→ Affinity of chemical groups: Affinity Chromatography
→ Surface Charge: Ion Exchange Chromatography
→ Hydrophobic Regions: Hydrophobic Interaction Chromatography
→ Size: Size Exclusion Chromatography (gel)

Basis of liquid Chromatography: Difference in affinity for column material

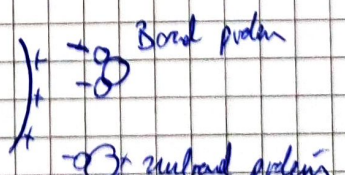
Affinity and Ion Exchange Chromatography

Affinity Chromatography:



→ highly selective
→ requires affinity tag

Ion Exchange Chromatography:



→ no affinity tag needed
→ high flow rates, high throughput

Biological Nanotechnology - 20 dtl 1

Protein Purification with Chromatography

Size Exclusion Chromatography (SEC): filter

-> high purity & polishing step

-> impurities can stick to column

Preparation, Extraction, clarification → Capture → Intermediate purification → polishing

General:

Trade-Off between: Speed, Resolution, Recovery, Capacity

=> Several Steps Required (multiple methods combined)

Fast Protein Liquid Chromatography (FPLC)

-> High level of automation: sample injection, flow, gradient setup, sample collecti

-> Online UV-Vis spectroscopy

eg:
Overexpression in E. coli => Pre-Purification => FPLC

Conclusion

-> Molecular cloning is powerful tool to generate constructs for protein production

-> Protein Production requires optimization, eg: temp during growth, cell density, concentration of inducer

-> Chromatography Methods enable fast and efficient purification of protein of interest

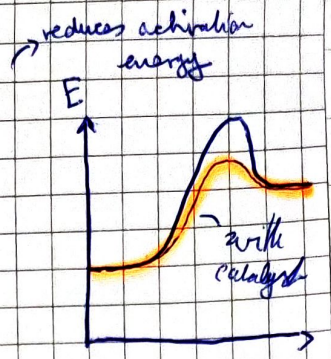
& several Steps needed

-> Protein Stability and solubility are important, improved by suitable buffer condition (pH, salt)

Biotechnische Verfahrenstechnik - Woche 2

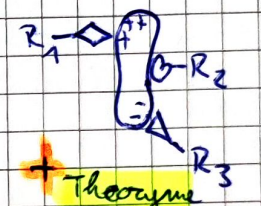
Enzyme Catalysis

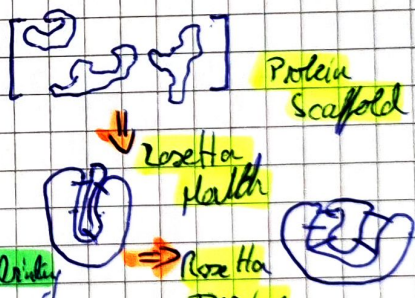
-> Catalyst lowers activation energy for reaction
-> Active Site: Special arrangement of amino acids
-> Substrate binds to active site of enzyme
-> Enzymes: high turnover numbers
   ↳ max number of substrate molecules converted to products by an enzyme molecule per unit time when enzyme is fully saturated with substrate
-> Suffix: -ase
-> digestive enzymes, eg: catalyze hydrolysis
-> eg: proteases hydrolyze peptide bonds



Computational Enzyme Design

-> Idealized model of desired active site based on quantum mechanically calculated structure of transition state
-> Docked into existing protein scaffolds, side chains need to placed
-> Binding pocket repacked computationally
-> Requirements:
   ↳ Suitable Theozyme (Theoretically Enzyme)
   ↳ (Crystal) Structure of scaffold
   ↳ Experimental testing of designs

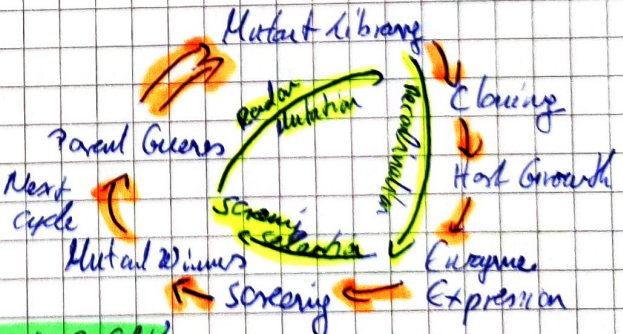




eg: -> Enzyme designed with Rosetta to catalyze cleavage of aldol product
-> Inhibitor: blocks active site, allows to image activity

Directed Evolution

1) -> Random Mutagenesis
2) -> Protein Production
3) -> Assay of Screening & selection



What factors can influence the protein production in E. Coli?
-> Genetic Factors: Plasmid Design, Host Strain
   ↳ Promoter Strength
-> Growth Conditions: Temperature, pH & Aeration, Induction Strength

Bioinformatik Vorlesung – Woche 2

Non-Natural Reaction: Diels-Alderase

Diels-Alder Reaction:

-> Enzyme Design with Rosetta

-> Improvement of the enzyme with directed evolution

Rosetta: Computational Protein and Enzyme Design

-> Multiple Applications for Rosetta: See Lec 2, Slide 17

-> CASP: Critical Assessment of techniques for Protein Structure Prediction

↳ tournament

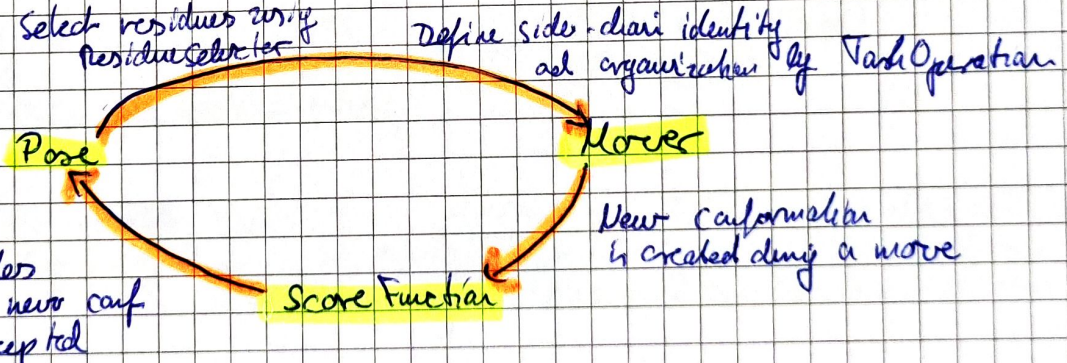
↳ Rosetta, winning with deep learning

Typically Rosetta Protocol

1) Start from conformation of biomolecule (pose)

2) Pose is altered via a Mover (detail/structure)

3) Conformation is evaluated by a Score Function



Which do you need to know about a reaction to design an enzyme?

-> Substrate & Products

-> Reaction Mechanism & transition state

-> Active site structure

-> Substrate Binding

Bioinformatik - Woche 2

Protein Containers

Interior

(encapsulates other molecules)

(= Protein based structure)

Protein Cage

Interface

(deliver drugs)

Exterior

Symmetric:
- → Structural & Robust against mutations
- → Stability
- → Efficiently Assembly: reliable
- → Functional Efficiently
- → Energetic Favorability
- → results from subunits efficient
- → functional groups, encapsulate

eg: Cowpea 31 nm, ~~[redacted]~~ (measure virus)

GFP = Green Fluorescent Protein

Container Engineering: Rational Design

n × [ ] + [ ] → [ ]

AaLS-neg
negatively charged host protein

GFP-R10
guest protein target with positive charges

Carged GFP-R10:
host (AaLS-neg) plus guest

~~[redacted]~~
Container Engineering: Specialized Containers / Carriers at the molecular or nanoscale level to encapsulate and deliver specific cargo for various applications in medicine.

Rational Design: Using existing knowledge of protein structure and function to design containers with specific properties and functionalities through computational modeling and targeted modifications, aiming for precise control over structure and function

Biohybrid Nanostructures Wdh 2

Container Engineering: Directed Evolution

Host can be improved by directed evolution

Mutation
selection
amplification

→ AaLS-13

Caged HIV protease (host plus guest)

Directed Evolution: Iterative rounds of mutation and selection to evolve containers with desired properties by introducing genetic diversity into a population of containers and selecting those with improved function.

Surface Charging of Proteins: GFP

GFP(-30) st GFP GFP(+36) GFP(+48)

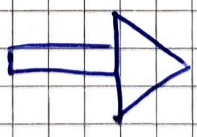
→ increased thermal stability and refolding (than uncharged)

→ uptake by mammalian cells (& eukaryotic cells)

→ Av NAPSA: Average Number of Neighboring Atoms (<10 A°) Per Sidechain Atom

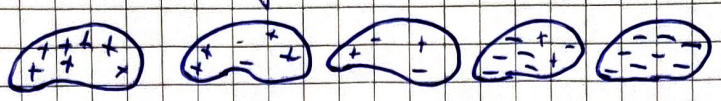
→ Low Av NAPSA: solvent-exposed residue

→ supercharged variants remain soluble when heated to 100°C and recover fluorescence after cooling



Biological Nanostructures - WdH 2

Surface Charging of Antibodies



Single-Chain Antibody

→ Increased Thermal Stability

→ Rosetta software for surface redesign (+, −, 0)

$$\text{Total Energy} = \sum \left(\begin{matrix} \text{Van der} \\ \text{Waals} \end{matrix} + \begin{matrix} \text{Electro-} \\ \text{statics} \end{matrix} + \text{Solvation} + \begin{matrix} \text{Hydrogen} \\ \text{Bonds} \end{matrix} + \text{Reference} \right)$$

"Unique cost for designing in each residue type"

Fernández: Surface charging with Rosetta

→ Fixbb module: keep backbone fixed, substitute side chains

↳ Allow native AA and Arg/Lys OR Glu/Asp

Extension to Protein Containers:

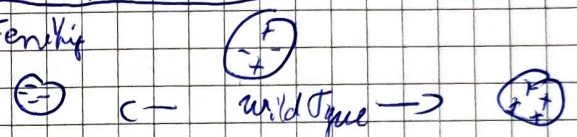
→ Symmetric Container

→ Only residues on outer surface

→ No contact with other monomers

→ Manual inspection

Fernández



wild type

with Rosetta software

Difference between the design of supercharged GFP and the design of the surface-charged antibody:

~~Surface Charging of Antibodies & GFPs~~

Design of supercharged GFP: enhance solubility, stability & potentially the intracellular delivery of GFP

~~(Directed evolution)~~

→ introduce high # of charged residues uniformly across the protein surface

design of the surface-charged antibody: focused on optimizing binding affinity, specificity, solubility and pharmacokinetics

→ more targeted modifications often in specific regions

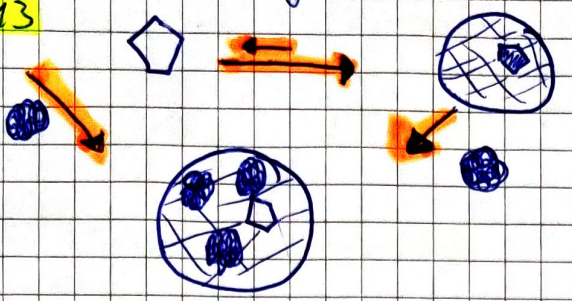
⇒ Reference energy for residues: (Antibodies)

↳ used in GFP

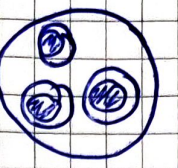
Biological Nanotechnology - Week 2

Morphological - Type Icosahedron

AaLS-13 Fragments



AaLS-13 cage



Fbu (p3)

Design of artificial protein cabanes

-> Redesign of contact interfaces

-> Tetrahedral (12 monomers) & octahedral (24 monomers) cabanes

-> Building Block & trimeric protein

Enzyme & Protein design

-> Computational methods are highly advanced and still improving

-> Experimental testing still necessary

- Enzymes:
  - -> Functional Design: Enzyme catalyzes (novel) reaction
  - -> Accuracy is important, chemical bonds need to be formed
  - -> Combination of computational design and directed evolution highly successful
- Protein Design:
  - -> Structural Design
  - -> Accuracy not as important, structural elements are connected
  - -> Novel assemblies can be generated

Bioinformatics Workbook — Week 3

Protein crystallography: Crystallization and crystal symmetry

Experimental determination of protein structures

Nuclear magnetic resonance (NMR):

- → in solution
- → Structure Solution Time-Consuming
- → Size limitation

Cryogenic Electron Microscopy:

- → frozen in water
- → large complexes
- → low resolution

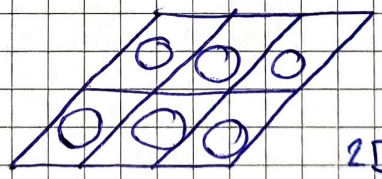
Protein Crystallography:

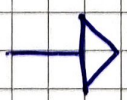
- → No size limitation ✓
- → High resolution ✓
- → Stable ✓

Protein Crystals

Protein Crystals:

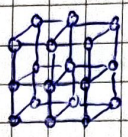
- → folded proteins
- → regular arrangement in three dimensions
- → interactions: H-bonds, electrostatic interactions and VdW-bonds, but all are weak
- → usually fragile
- → large amount of solvent





Solid State Nanostructures - Week 3

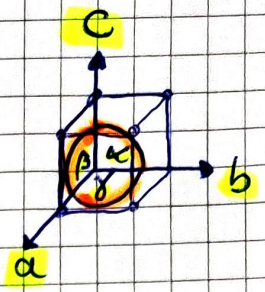
Crystal Lattice and Crystal Systems



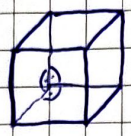
-> Crystal lattice constructed from unit cells

-> Translational Symmetry

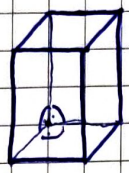
-> Seven Crystal systems, depends on geometry of unit cell



Cubic:
$a = b = c$
$\alpha = \beta = \gamma = 90°$

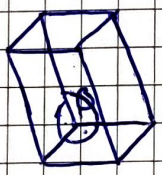


Tetragonal:
$a = b \neq c$
$\alpha = \beta = \gamma = 90°$

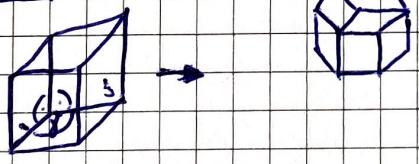


Orthorhombic:
$a \neq b \neq c$
$\alpha = \beta = \gamma = 90°$

Monoclinic:
$a \neq b \neq c$
$\alpha = \gamma = 90° \neq \beta$

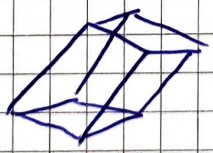


Hexagonal:
$a = b \neq c$
$\alpha = \beta = 90°, \gamma = 120°$



Rhombohedral:
$a = b = c$
$\alpha = \beta = \gamma \neq 90°$

Triclinic
$a \neq b \neq c$
$\alpha \neq \beta \neq \gamma \neq 90°$



Bändyhtid Manchuku 2dff 3

Bravais lattices

→ Kattice defined by ea vector triple

→ 14 Bravais Lattice with centering:

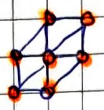
P : Primitive
I : Body centered
C : Base centered
F : Face centered

Bravais Kattice:
the lattice in which the atoms or all the atoms at kattice points are identical or if the surroundings of each lattice points are identical

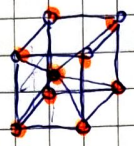
↳ symmetric by Translation

Cubic:

P:

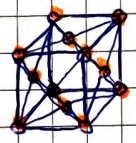


I:



C:

F:



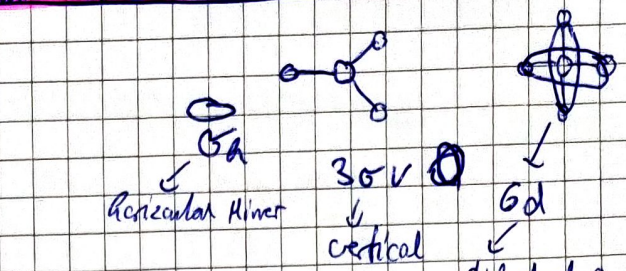
Bodylad Vorbereitung Teil 3

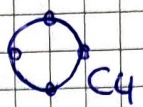
Symmetry Elements

Schönflies:

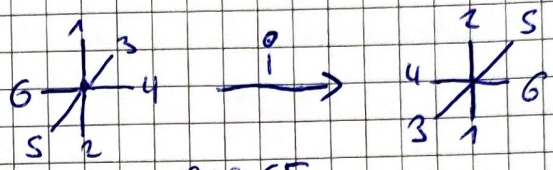
Point Symmetry:

- **Identity:** E
- **Rotation axes:** C_n, n-fold rotation axis
- **Mirror Planes:** σ_h / σ_v / σ_d, mirrors
- **Inversion Center:** i
- **Rotoinversion:** —
- **Rotary reflection axis:** S_n → & Rotation Reflexion
- **Space Symmetry:**
- **Glide Plane &:** —
- **Screw Axes:** —



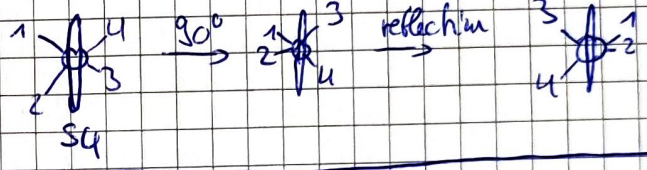


x · σ;



eg: SF_6

$n: 360°/n$



Hermann-Mauguin:

n/m → if rotation & mirror plane have same direction

Point Symmetry:

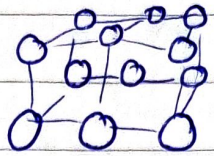
- **Identity:** 1
- **Rotation Axes &:** n — $\frac{360°}{n}$
- **Mirror Planes:** m → Spiegelebene — 3m (3 MP in different directions), mm (2 Spiegelebenen senkrecht)
- **Inversion Center:** $\bar{1}$ — same as above
- **Rotoinversion &:** $\bar{3}$, $\bar{6}$, etc. → Drehung um $\frac{360°}{n}$ & Punkt-Spiegelung
  - ↳ $\bar{n}$ = n/1
- **Rotary Reflection axes:** —

Space Symmetry:

- **Glide Plane:** n, d, a, b, c → reflection combined with translation
- **Screw axes:** 2_1, 3_1, etc → rotation & translation
  - ↳ Rotate by $\frac{360°}{2}$ & translate by 1/2
  - ↳ Rotate by $\frac{360°}{4}$ & translate by 1/4

Biological Macromolecules Week 3

Why are there no glide planes or roto-inversion elements found in protein crystals?



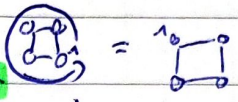
→ Chirality of AA's

↳ building blocks of Proteins are L- / D- AA's

↳ natural protein consists of only L-AA's

↳ protein itself is asymmetric

→ only translation → but no mirror or rotation

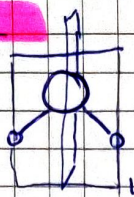


↳ L-!

Bildung der Vorlesung Wdh #3

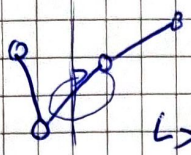
Point Groups

eg:

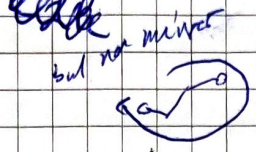


C_2

↳ C_{2v} point group.



↳ C_2



Space Groups:

Point Group: Finite Object → Schönflies

Lattice: Translational Symmetry

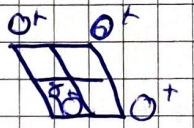
Point Groups + Lattice Translations: Space Group

Space Groups: Symmetry of the crystal

Unit Cell Centering + Symmetry Elements = Space group symbol

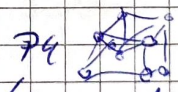
(P/C/I/F/R) a-Direction b-Direction c-Direction

↳ only 1 Rotation Symmetry or screw Axes!



P1 1) x, y, z

↳



P4

↳ no redundant

P222

Asymmetric Unit:

- → Smallest building unit of the unit cell
- → Does not contain any symmetry
- → Asymmetric unit + Symmetry Operators = Contents of the unit cell

Equivalent Positions:

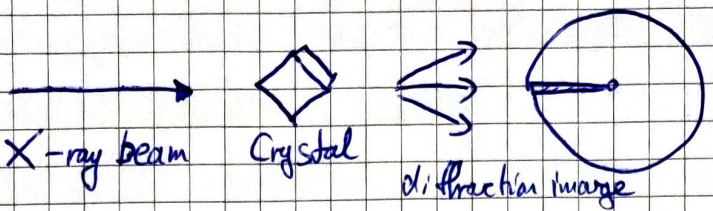
- → Symmetry-related sites in unit cell
- → General equivalent positions
- → Special positions: required object of certain symmetry

What is the size of the asymmetric unit for space group P1? P-1?

P1: entire unit cell ; P-1: half of the unit cell

Biophysik Vorlesung Woche 3

Basics of X-ray diffraction



→ Single molecule: weak interaction with X-rays

→ Crystal: Collective interaction with X-rays

→ X-rays: Wavelength is similar to the features of molecules

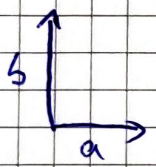
→ X-rays interact with electrons of the atoms inside the crystal

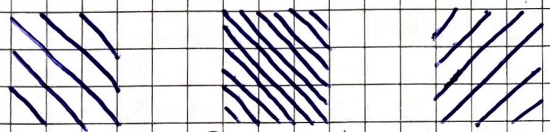
Radiation for diffraction

→ X-rays
→ Electrons
→ Neutrons

Goal: Determination of Atomic Coordinates

Lattice Planes and Miller Indices





(1 1 0) planes (1 3 0) planes (-2 1 0) planes

↳ more often cuts "b"

Miller Index: Intercept with the reciprocal value of the index

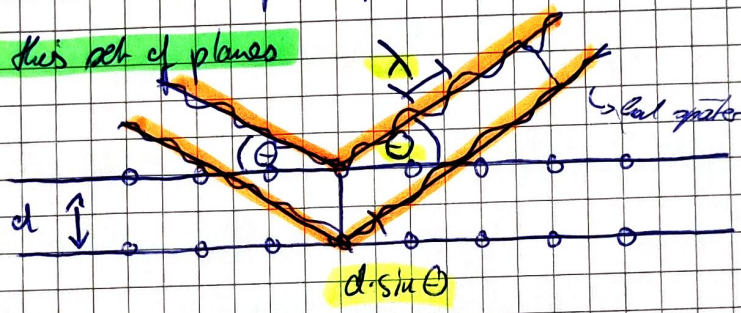
eg. plane with index 2 intercepts at 1/2

Lattice Planes: divide the crystal into repeating units and are described by their orientation relative to the crystal lattice axes.

Röntgenstrahl Vorlesung Woche 2

Bragg's Law

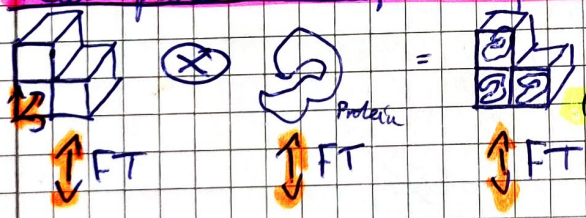
- Distances between atoms and wavelength have similar dimensions
- diffraction at a grating of atoms
- Simplification: Reflection on a fictitious set of planes that go through the crystal
- Atoms does not need to be on planes
- If the incident angle and the distance between planes result in a phase shift of the two waves of multiples of the wavelength constructive interference
- Bragg Equation $2d \cdot \sin\Theta = n \cdot \lambda$ (d: spacing between crystal lattice planes, Θ: angle of incident, λ: wavelength)
- Miller Indices hkl of the planes, corresponds to the reflection hkl from this set of planes



↳ wie aus in gerader Linie wird reflektiert

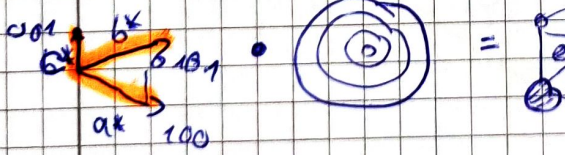
↳ hat später an

Real Space & Reciprocal Space



Convolution: Combination of two functions

Real Space



Fourier Transform: Relation between two domains
eg: time and frequency

Reciprocal space

3dyl1de Denaturation Wdh 3

From reciprocal space to electron density

Reciprocal Space:

Reflections on the detector

Structure factor F of a reflection hkl (h):

$$F_h = \sum_{j=1}^{atoms} f_j^B \exp(2\pi i\, h\, x_j)$$, with atomic scattering factors f and atom positions x

Fourier Transform into Real Space: with $F_h = |F_h| \cdot \exp(i\,\phi_h)$

$$\rho(x) = \frac{1}{V} \sum_h |F_h| \exp(i\phi_h) \exp(-2\pi i\, hx)$$

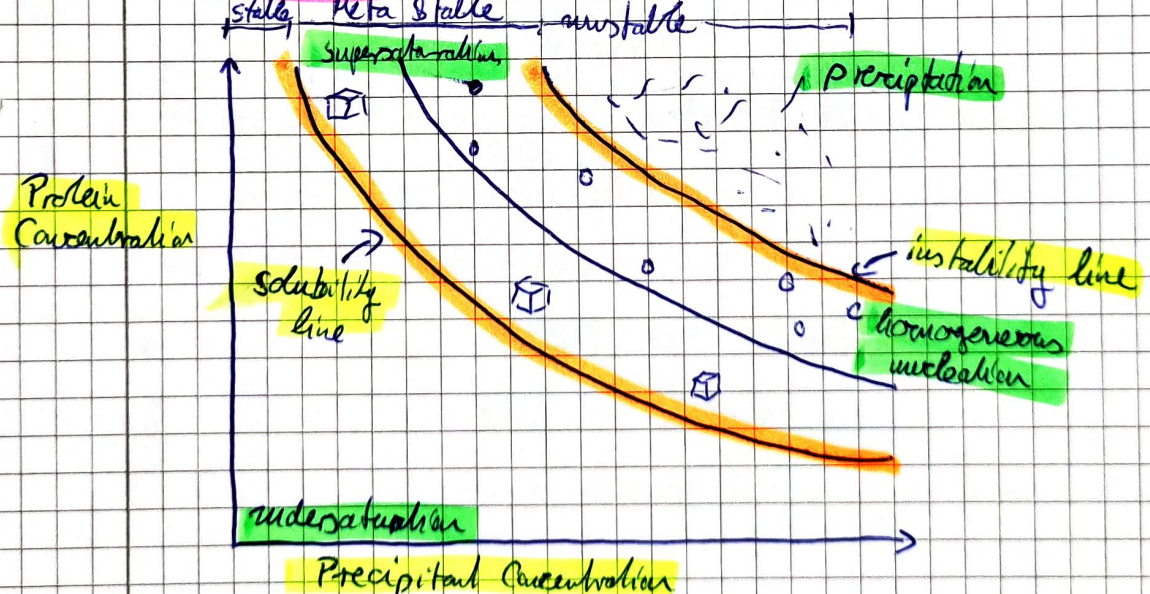
From electron density to model

→ Electron density inside crystal

Model: Peptide chain that fits the electron density

Goal: Exact 3D model of the protein

Phase Diagram of Protein Crystallization



↳ are agents added to the solution to reduce the solubility of the protein, thereby promoting crystallization

3/Lysinob Vorlesung Diff 3

Chaotropic and Kosmotropic Ions

Chaotropic Ions prefer the weakly H-bonded surface

→ large, singly charged ions → disturbs the structure of water

→ low charge density

→ weakly hydrated

eg: SCN^-

Protein: Increase Denaturation
↳ unfolding
↳ Salting in

Kosmotropic Ions prefer full hydration in the bulk

→ small / multiply charged ions

→ high charge density

→ strong interaction with water → enhance the structure of water

→ strongly hydrated

eg: SO_4^{2-}

Protein: Increase Stability
↳ folding
↳ Salting out

Solubility of Proteins

Ions can affect protein in two ways:

→ Ions can stabilize or destabilize the protein structure:
Kosmotropic Chaotropic
stability vs denaturation

→ Ions can increase or decrease solubility of proteins:
Chaotropic Kosmotropic
Salting In vs Salting Out

Protein Stabilization and Denaturation

Ion binding to proteins: • weakly hydrated anions, eg: SCN^- bind strongly to amide backbone of proteins.

• strongly hydrated anions, eg: Ca^{2+} bind strongly to charged side chains

Chaotropic ions eg: SCN^- and Guanidinium:

• enhance unfolding of proteins

• Urea: inserts into hydrophobic pockets

• Related to hydrophobic effect: protein can unfolded

Bottle Mar 28th 3

Protein Solubility: Salting In and Salting Out

Salting In:

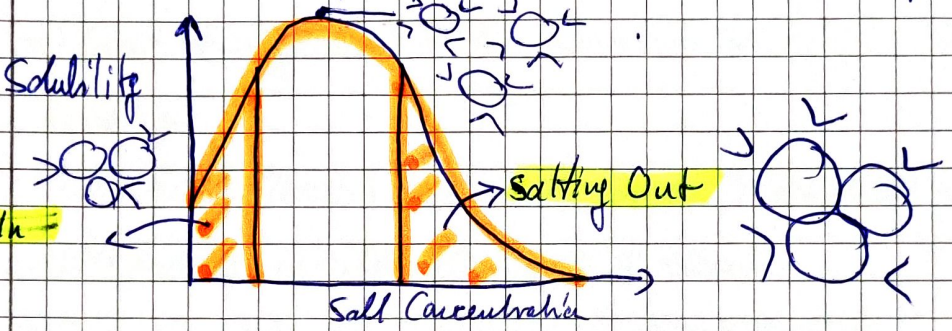
→ Ions coordinate to protein molecules, either equip them with charge for solubility, or break up electrostatic interaction between proteins by shielding

Salting Out:

→ Ions that strongly interact with water structure, eg: Sulfate ions, remove water layer around proteins

→ exposes hydrophobic patches on proteins, induces aggregation

→ ammonium sulfate precipitation is a standard step in protein purification



Protein Solubility and Crystallization

Salts: salting out / salting in

Organic additives such as glycerol: Change the dielectric constant of water

Polymers such as polyethylene glycol (PEG):

increase molecular crowding, reduces solubility

Protein charges affect solubility:

isoelectric point (pI): pH value where protein carries no net charge

Biophysical Macromolecular with 3

Crystallization: High Throughput Screening

Automisation of finding good parameters using Pipetting Robot.

Aqueous Solution.

Parameters:
- Protein Concentration
- pH value
- Salt Concentration
- Precipitation Agent
- Drop Size

Methods:
- Hanging Drop
- Sitting Drop

Goals: Reproducible Conditions, Highly-Ordered Crystals

Conclusion

→ Space Groups consists of Bravais lattice centering and symmetry operators

→ Asymmetric Unit plus symmetry operators generate the unit cells contents

→ Diffraction: Reflection from a set of planes

→ Fourier Transform from reciprocal space to real space

→ Proteins can be crystallized using high throughput approaches

→ Precipitant (salt, other additives) can control solubility of proteins

Why is it important to test several hundred conditions for the crystallization?

↳ reproducible conditions

↳ Stochastic increase for same result with same conditions

↳ discover unexpected conditions ⇒ Optimize for all crystals